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2025 QATS Chem exam - practice exam

Chemistry (MacRobertson Girls High School)

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Name: _____

Chemistry Units 3 & 4

Question and Answer Book

VCE ® Unit 3 & 4 Practice Written Examination

- **Reading time:** 15 minutes
- **Writing time:** 2 hours and 30 minutes

Approved materials
- Pens, pencils, highlighters, erasers, sharpeners and rulers
- One scientific calculator

Materials supplied
- Question and answer book of 29 pages
- Multiple-Choice Answer Sheet

Instructions
- Write your **student name** in the space provided above on this page.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book.

Students are **not** permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

Contents pages

Section A (30 questions, 30 marks) _____ 2–11
Section B (9 questions, 90 marks) _____ 12–29

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Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on the Multiple-Choice Answer Sheet.
- Choose the response that is **correct** or that **best answers** the question.
- A correct answer scores 1, an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1

Which of the following statements about the fermentation of glucose is correct?

A. It has a 100% atom economy percentage.

B. It is an exothermic process.

C. Water is a reactant.

D. Oxygen gas is a product.

Question 2

The respective number of carbon and hydrogen environments produced in NMR spectra by the compound propan-1-ol is

A. 3 and 3

B. 4 and 3

C. 3 and 4

D. 4 and 4

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Question 3

Consider the following reaction involving the production of nitrogen dioxide.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The reaction is endothermic.

To maximise the yield of nitrogen dioxide, the experiment should be done under

A. high pressure and high temperature.

B. high pressure and low temperature.

C. low pressure and high temperature.

D. low pressure and low temperature.

Question 4

The correct IUPAC name for the ester produced from the reaction between butan-1-ol and propanoic acid is

A. propyl butanoate.

B. butyl propanoic acid.

C. ethyl pentanoate.

D. butyl propanoate.

Question 5

The mass of cobalt from an $CoCl_2$ solution that could be deposited on a carbon electrode if a current of 4.50 A was applied for 20.0 minutes is closest to

A. 0.0540 g

B. 1.65 g

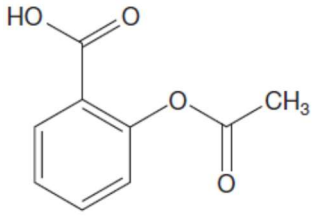
C. 3.30 g

D. 0.0270 g

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Question 6

The structure of aspirin is shown below.



The functional groups present include

A. carboxyl and aldehyde.

B. hydroxyl and amino.

C. carboxyl and ester.

D. hydroxyl and amide.

Question 7

A solution calorimeter was calibrated via electrical calibration. A voltage of 4.00 V, using a current of 2.00 A was applied for 5.00 minutes. The change in temperature of the water was 2.60 °C. The calibration factor is

A. 15.4 J $°C^{-1}$

B. 923 J $°C^{-1}$

C. 104 J $°C^{-1}$

D. 6240 J $°C^{-1}$

Question 8

The number of structural isomers of C_4H_9Cl is

A. 2

B. 3

C. 4

D. 5

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Question 9

Consider the following statements about proteins

i. Denaturation of proteins involves the disruption of both the primary and secondary structures
ii. Enzymes can be classed as proteins
iii. All proteins contain sulphur

Regarding these statements, which is/are correct?

A. I only

B. II only

C. II and III

D. I, II and III

Question 10

When aqueous magnesium chloride is electrolysed, the product formed at the cathode is

A. hydrogen gas.

B. calcium metal.

C. chlorine gas.

D. oxygen gas.

Question 11

The systematic name of the compound $CH_3CHOHCHClCH_3$ is

A. 2-chlorobutan-3-ol.

B. 2-chlorobutanol.

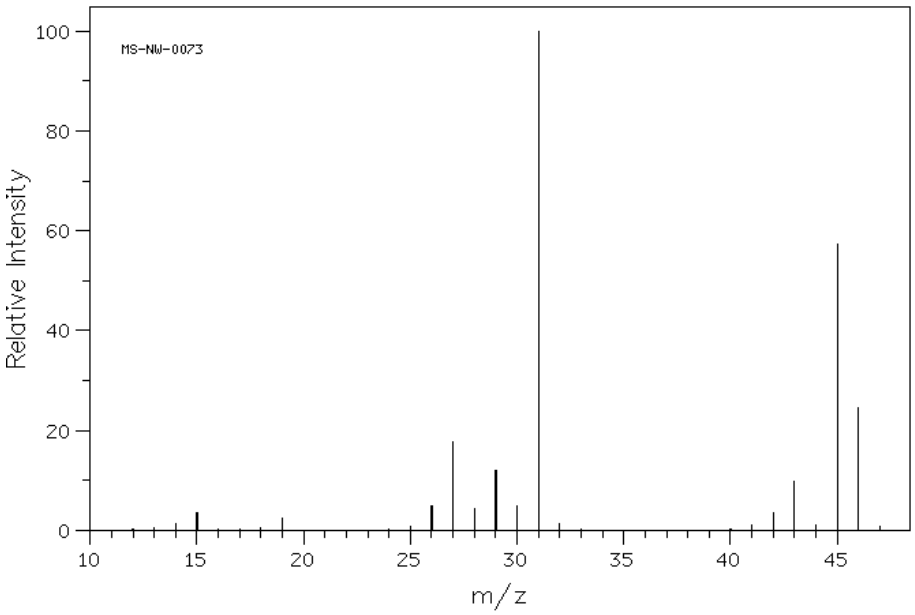
C. 3-chlorobutan-2-ol.

D. 3-chlorobutanol.

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Question 12

Consider the following mass spectrum for ethanol.



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National Institute of Advanced Industrial Science and Technology

The base peak has an m/z value of

A. 15

B. 31

C. 45

D. 46

Question 13

Which of the following compounds would be expected to have the highest boiling point?

A. propane

B. propanone

C. propanal

D. propan-1-ol

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Question 14

The reaction of acidified potassium permanganate with butan-2-ol is most likely to produce

A. an ester.

B. an alcohol.

C. a ketone.

D. an aldehyde.

Question 15

Which section of a scientific poster should data collected during the experiment be recorded in?

A. introduction

B. method

C. results

D. conclusion

Question 16

Which of the following statements about the rate of a chemical reaction is **incorrect**?

A. Only temperature can increase the rate of a reaction.

B. Powdered magnesium will react with hydrochloric acid faster than a strip of magnesium.

C. If the activation energy is lowered, the rate of reaction will increase.

D. Increasing the concentration of only one reactant will increase the rate of a reaction.

Question 17

Ammonia gas can be produced via the following equilibrium reaction

$$4NO(g) + 6H_2O(g) \rightleftharpoons 4NH_3(g) + 5O_2(g) \qquad \Delta H = +ve$$

If the temperature is increased at constant pressure, the value of K and the reaction rate will respectively

A. increase and increase.

B. decrease and increase.

C. increase and decrease.

D. remain the same and decrease.

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Use the following information to answer Questions 18-20

A galvanic cell is set up using the following two half-cells:

Half Cell 1: $Co^{2+}(aq)/Co(s)$. The electrode is cobalt metal.

Half Cell 2 $Fe^{2+}(aq)/Fe^{3+}(aq)$. The electrode is platinum metal.

The two half-cells are connected using a salt bridge containing $KNO_3(aq)$ and an external circuit to produce a voltage. The concentration of each solution is 1.0 M.

Question 18

The expected voltage produced by this cell is

A. 1.21 V

B. 0.16 V

C. 1.60 V

D. 1.05 V

Question 19

What option below best describes the operation of the galvanic cell?

A. The cobalt half-cell is the cathode and electrons flow to this half-cell.

B. The iron half-cell is the anode and electrons flow to this half-cell.

C. The iron half-cell is the cathode and electrons flow to this half-cell.

D. The cobalt half-cell is abode and electrons flow to this half-cell.

Question 20

The flow of ions in the salt bridge is best described by stating that

A. cations flow to the cathode and anions flow to the anode.

B. cations flow to the anode and anions flow to the cathode.

C. electrons flow to the anode and cations flow to the cathode.

D. electrons flow to the cathode and anions flow to the cathode.

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Question 21

A controlled variable is best described as a variable that is

A. altered in different trials.

B. measured.

C. kept constant.

D. always included in the aim of the experiment.

Question 22

When a catalyst is added to a system at chemical equilibrium

A. the yield of the products is increased.

B. the forward reaction rate increases only.

C. the reactants molecules have a greater energy to overcome the activation energy.

D. the activation energy is lowered.

Question 23

The oxidation number of the chromium atom in the dichromate ion $Cr_2O_7^{2-}$ is

A. $+3$

B. $+6$

C. $+12$

D. -6

Question 24

The digestion of carbohydrates in the human body can best be described as a

A. condensation reaction with water as a product.

B. hydrolytic reaction with water as a product.

C. hydrolytic reaction with water as a reactant.

D. condensation reaction with water as a reactant.

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Question 25

In a redox titration, the following values were obtained for the titres in each trial.

11.00 mL, 11.50 mL, 12.00 mL, 10.00 mL & 10.00 mL

The average value of these five trials was 10.90 mL. If the true value was 10.95 mL, what is the best description of the data?

A. accurate and precise

B. accurate but not precise

C. not accurate but precise

D. not accurate and not precise

Question 26

Consider the following reaction for the combustion of ethene.

$$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(g)$$

The reaction takes place at constant temperature and pressure.

If 100 mL of ethene reacts with 400 mL of oxygen gas, the total volume of gas present when the reaction has gone to completion is

A. 200 mL

B. 300 mL

C. 400 mL

D. 500 mL

Question 27

Methane can be sourced using two methods. The first method involves fracking to collect the gas produced from the decay of living plants and animals over millions of years. It can also be produced as biogas from the decomposition of plants and animals without the presence of oxygen. The Bunsen burners in a school laboratory use methane produced as biogas.

The green chemistry principle that this most aligns with is

A. catalysis.

B. design for degradation.

C. use of renewable feedstocks.

D. economic prosperity.

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Question 28

Pentanoic acid and ethyl propanoate both have the same molecular formula. The two compounds were analysed using IR spectroscopy and two IR spectra were produced. The absorption band, in cm–1, which would best allow the two compounds to be identified is

A. 1050–1410

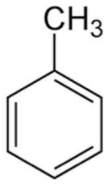
B. 2850–3090

C. 2500–3500

D. 3300–3500

Question 29

The structure of the compound toluene is shown below



The presence of the benzene ring means that this compound is considered highly non-polar. The best substance to extract this compound would be

A. water.

B. ethanol.

C. hexane.

D. ethanoic acid.

Question 30

The best description of how a competitive inhibitor works to disrupt the function of an enzyme is

A. it binds to the active site of the enzyme.

B. it denatures the enzyme.

C. it binds to the substrate.

D. it chemically reacts with the enzyme.

End of Section A

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Section B

Instructions

- Answer all questions in the space provided.
- Write your responses in English.
- Give simplified answers to all numerical questions, with an appropriate number of significant figures; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions; no marks will be given for an incorrect answer unless it is accompanied by details of the working.
- Ensure that chemical equations are balanced and that the formulas for individual substances include an indication of state, for example, $H_2(g)$, NaCl(s).
- Unless otherwise indicated, the diagrams in this book are not drawn to scale.

Question 1 (10 marks)

Consider the food label shown below.

	Amount per 100 g
Proteins	3.4 g
Carbohydrates	24.6 g
Fats	0.30 g

a. Calculate the amount of energy provided in a 100 g sample of this food. 2 marks

b. Glucose is a monosaccharide that is often referred to as a simple carbohydrate.

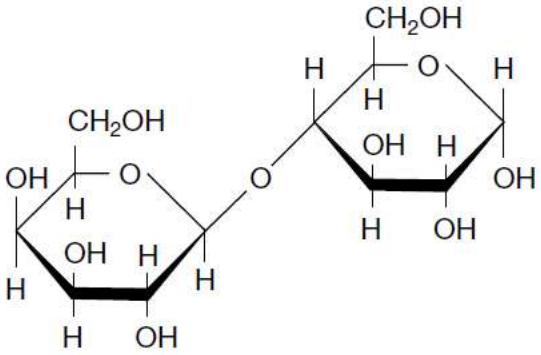
Write the balanced chemical reaction for respiration. 2 marks

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c. Glucose can be provided in our diet through a range of sources. One such source is lactose. Lactose is a disaccharide made up of a glucose molecule and a galactose molecule.

Name the reaction that breaks down lactose into glucose and galactose. 1 mark

d. Shown below is the structure of lactose. On the structure, circle the link that joins the two monosaccharides and name this link. 2 marks



e. When our body has an excess supply of glucose, it can convert it into a large molecule that is usually stored in the liver. Name this molecule. 1 mark

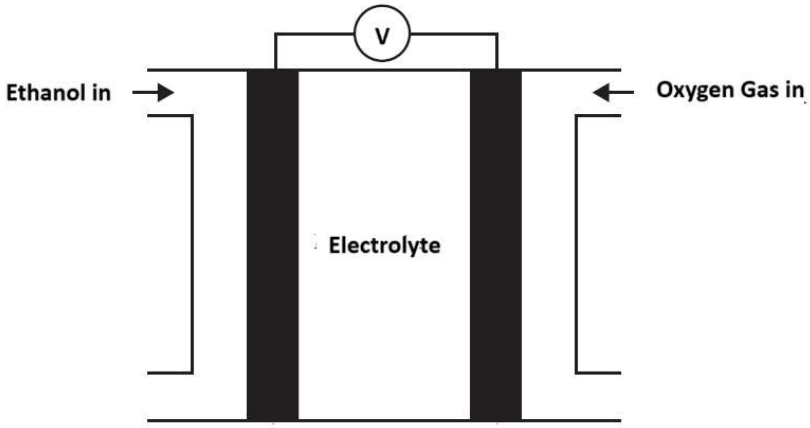
f. Starch is a complex carbohydrate that is found in the fleshy part of fruits and vegetables. The structure of starch is different from the structure of the compound named in **Part e.**

State a key difference between the structures and explain how this can affect the digestion of the molecules. 2 marks

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Question 2 (12 marks)

A schematic diagram for an ethanol fuel cell is shown below.



The aqueous electrolyte contains potassium hydroxide (KOH).

The half equation that represents the reaction occurring at the cathode is shown below.

$$O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$$

a. Identify the change in oxidation number of the oxygen atom shown above. 1 mark

b. State the polarity of the cathode in this fuel cell. 1 mark

c. Write the half equation that represents the reaction occurring at the anode. 2 marks

d. Hence, write the overall equation for this fuel cell. 1 mark

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e. Explain what is meant by the term porous and why the electrodes above have this feature. 2 marks

f. With reference to the green chemistry principle of renewable feedstocks, explain how the ethanol used in this cell could be produced. 2 marks

g. The fuel cell operated for 24.0 hours and delivered a current of 10.0 A. Calculate the mass of ethanol that was required during this time. 3 marks

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Question 3 (9 marks)

Consider the equation for the production of sulphur trioxide shown below.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -198\ \text{kJ mol}^{-1}$$

a. Write the expression for the equilibrium constant. 1 mark

b. Calculate the value of K given the following concentrations when equilibrium is reached at 800 K. $[SO_3] = 0.500\ \text{M}\ \ [SO_2] = 0.070\ \text{M}\ \ [O_2] = 0.275\ \text{M}$. 2 marks

c. Once the reaction has reached equilibrium, what changes could be made to both temperature and volume to increase the yield of SO_3? Explain how these changes will maximise the yield. 4 marks

d. A catalyst is often used in this reaction. Explain what effect a catalyst has on the rate of reaction and why catalysts are often used in exothermic reactions. 2 marks

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Question 4 (13 marks)

Ethanol can be produced via the following organic reaction pathway.

$$CH_3CH_2Cl + NaOH \rightarrow CH_3CH_2OH + NaCl$$

a. Write down the name of this type of reaction. 1 mark

b. Calculate the percentage atom economy to produce ethanol via this organic reaction pathway. 2 marks

c. An alternative method to produce ethanol is an addition reaction using ethene and water. Write down the complete balanced chemical equation for this method. 2 marks

d. State a simple laboratory test that could be performed to determine if an alcohol was produced. Explain how you could determine if an alcohol was produced via your stated test. 2 marks

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e. Consider the following chemical equation to produce ethyl propanoate.

$$CH_3CH_2OH(l) + CH_3CH_2COOH(l) \rightarrow CH_3CH_2COOCH_2CH_3(l) + H_2O(l)$$

i. If 2.40 g of ethanol was reacted with excess propanoic acid, calculate the theoretical amount of ethyl propanoate that should be produced. 2 marks

ii. Calculate the percentage yield of the experiment if it was determined that 4.65 g of ethyl propanoate was produced. 1 mark

f. The ethyl propanoate was purified and laboratory analysis of its melting point was undertaken.

i. Identify **one** process that would be needed to purify the ethyl propanoate. 1 mark

ii. Explain how the results of this analysis would be able to determine if the purification process was successful. 2 marks

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Question 5 (10 marks)

A student in a Year 12 Chemistry class used a spirit burner to calculate the molar heat of combustion for ethanol. Below is a table showing their results. It was done at SLC.

Mass of water used	200.0 g
Volume of Ethanol	2.00 mL
Density of Ethanol	0.785 g mL^{-1}
Temperature increase of water	14.2 °C

a. Calculate the energy absorbed by the water. 1 mark

b. Assuming all the energy absorbed from the water came from the combustion of ethanol, what is the calculated value for the enthalpy change in kJ mol^{-1}? 2 marks

c. Using the theoretical value for Δ*H* (ethanol) in the Databook, calculate the percentage of energy that was lost to the environment. 2 marks

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In an alternative experiment, a solution calorimeter was used to determine the enthalpy change for the following process.

$$\mathrm{HNO_3(l)} \rightarrow \mathrm{HNO_3(aq)}$$

The first step was to calibrate the solution calorimeter. An electric current of 3.00 A with a voltage of 6.0 V was passed through 150 mL of water for 60 s. The change in temperature was 1.2 °C

d. Determine the calibration factor. 2 marks

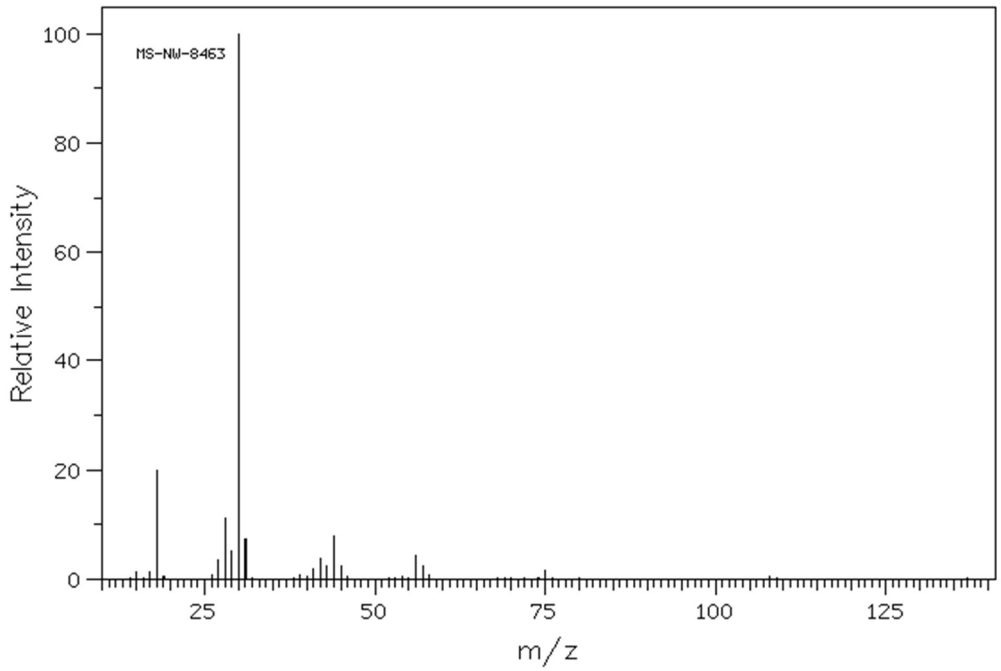
e. If 1.50 g of nitric acid was placed in 150 mL of water in the same solution calorimeter, determine the Δ*H* of the above process if the temperature increased from 22.5 °C to 24.0 °C. 2 marks

f. Multiple trials were conducted, and it was found that the calculated Δ*H* value was consistently less than the theoretical value. Identify **one** reason why this would be the case. 1 mark

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Question 6 (9 marks)

A compound with an empirical formula of C_3H_9NO produced a mass spectrum as shown below.



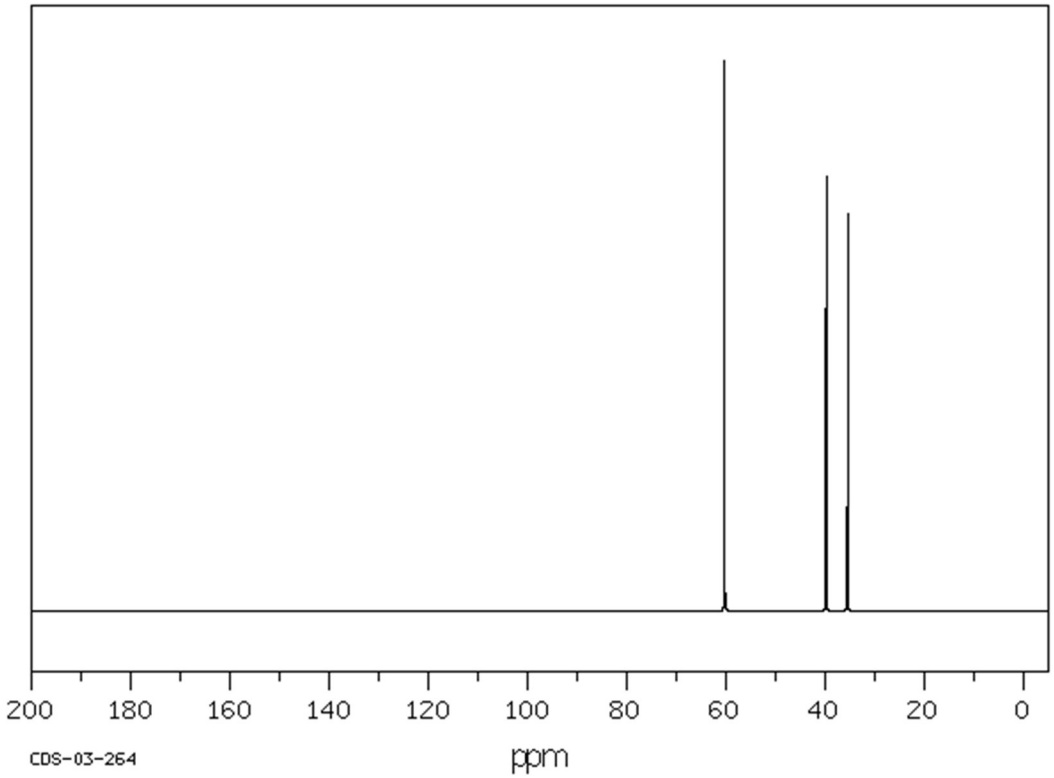
Data: SDBS Web, <http://sdbs.db.aist.go.jp>,
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a. The molecular ion peak has an m/z value of 75. Determine the molecular formula of the compound. 1 mark

b. Identify the value of the base peak and write down a possible fragment that would produce this peak. 2 marks

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The carbon-13 NMR produced by the compound is shown below.



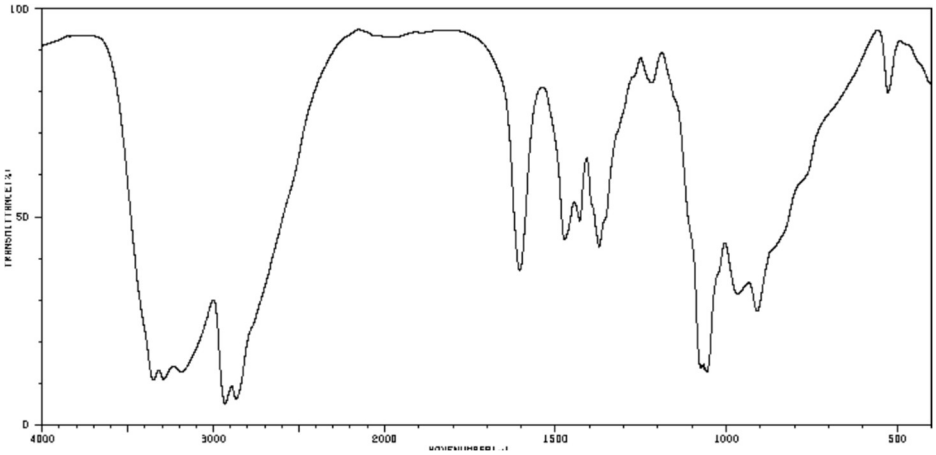
Data: SDBS Web, <http://sdbs.db.aist.go.jp>,
National Institute of Advanced Industrial Science and Technology

c. Using information provided in the spectrum above, determine how many carbon environments are present in the molecule. Justify your response. 2 marks

d. The first peak in the NMR above occurs at approximately 35 ppm. Using the Databook, write down a possible structure that would produce this peak. 1 mark

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The IR spectrum produced by the compound is shown below.



Data: SDBS Web, <http://sdbs.db.aist.go.jp>,
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e. An amino group is present. Using the spectrum above, identify the other functional group present. 1 mark

The low-resolution H-NMR produced by the compound showed 5 individual peaks at the following values.

Peak	δ (ppm)
1	1.7
2	2.6
3	2.9
4	3.7
5	7.4

f. Using the above information, draw a possible skeletal formula for the compound in the space below. 2 marks

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Question 7 (11 marks)

Polymer electrolyte membrane cells are a method of producing hydrogen gas via electrolysis. One such method involves the use of solar energy to power the cell.

a. State the overall energy change in this cell that produces hydrogen gas. 1 mark

b. In the above design, the hydrogen produced can be referred to as green hydrogen. Explain what is meant by the term green hydrogen. 1 mark

The half equation representing the reaction at the cathode in acidic conditions is shown below.

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

c. Write the half equation for the reaction occurring at the anode. 1 mark

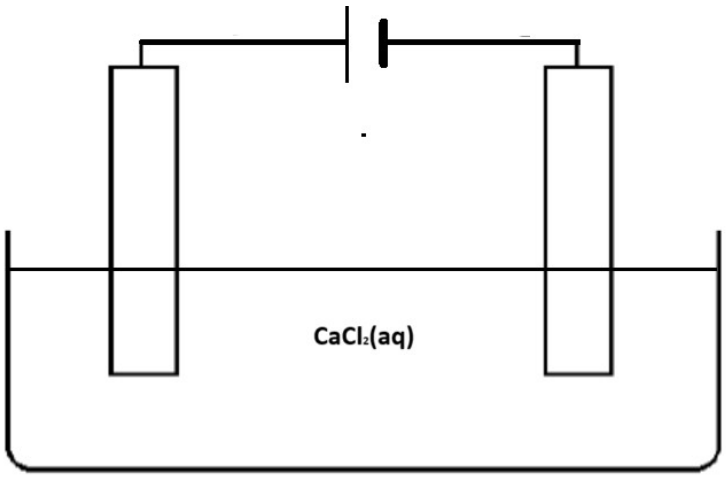
d. Hence, write the overall equation for this cell. 1 mark

Another method to produce green hydrogen is artificial photosynthesis.

e. State **one** thing in common between artificial photosynthesis and the photosynthesis that occurs in plants. 1 mark

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In a Year 12 Chemistry class, a solution of calcium chloride underwent electrolysis. A diagram of the cell is shown below. Both electrodes were made of carbon.



f. Write down the half equation for the reaction occurring at the cathode. 1 mark

g. What effect, if any, would occur if the anode was replaced with iron metal? Use an equation to help explain your response. 2 marks

Lithium batteries have become an important part of modern society. In several lithium batteries, the following half equation occurs at the anode.

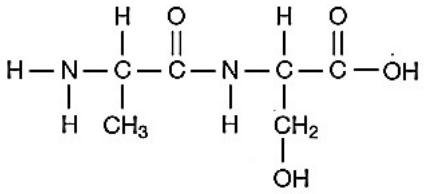
$$\mathrm{Li(s)} \rightarrow \mathrm{Li^+ + e^-}$$

h. Calculate the mass of lithium that would be required to produce a current of 5.0 A for a period of 6.00 hours. 3 marks

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Question 8 (8 marks)

Consider the dipeptide shown below.



a. Identify the **two** amino acids that make up this dipeptide. 2 marks

b. On the diagram above, circle a chiral carbon. 1 mark

c. Haemoglobin is a protein made of 574 individual amino acids. This results in a complex structure that allows the molecule to transport oxygen around the human body. Explain the difference in secondary and tertiary structure of this protein, referring to the intermolecular forces. 3 marks

d. Pepsin is another protein in the human body that helps digest proteins in the acidic environment of the stomach. Explain why pepsin loses its ability to digest proteins in a basic environment. 2 marks

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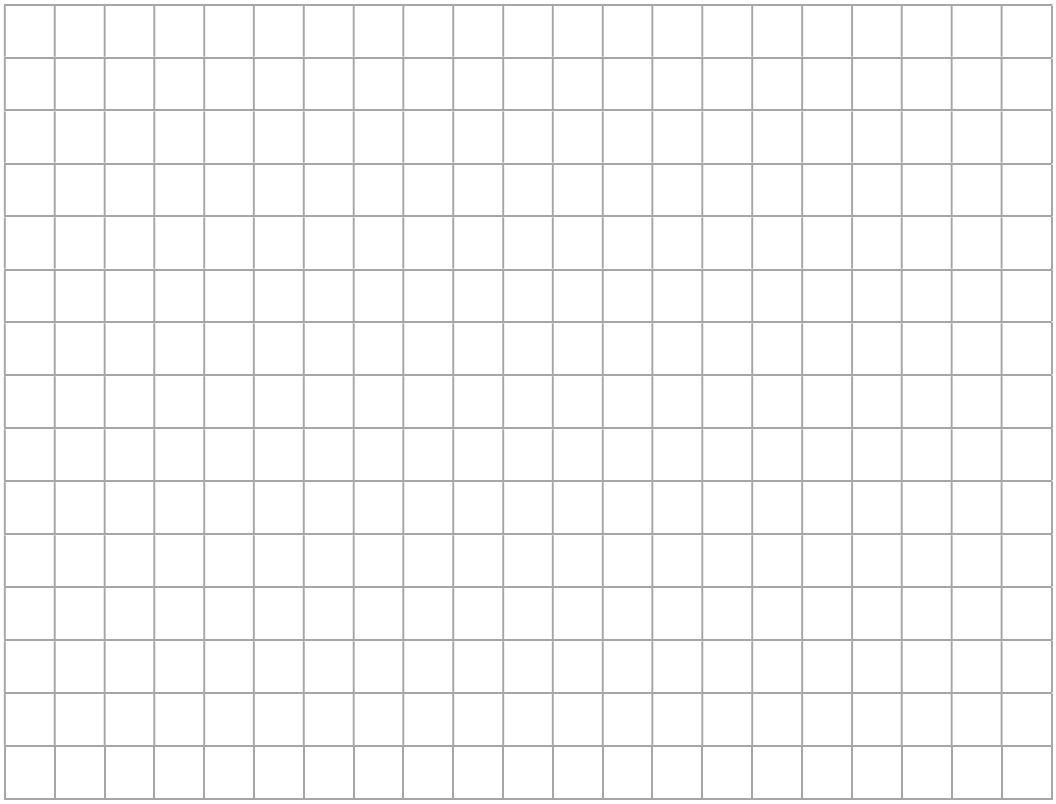
Question 9 (8 marks)

An experimental analysis was undertaken, using HPLC, to determine the concentration of caffeine in a sample of coffee.

The initial step in the process was to create a calibration curve from a set of standard solutions. The following results were obtained.

Concentration (g L^{-1})	Area under the peak
0.00	0
0.20	495
0.40	1002
0.60	1505
0.80	1997

a. Using the data above, plot a calibration curve. 2 marks



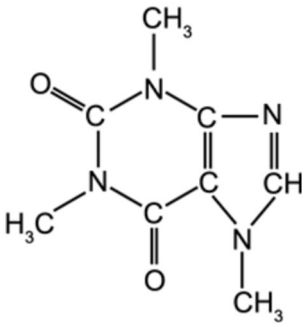
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b. State the independent variable for the initial step. 1 mark

c. The sample of coffee with an unknown concentration of caffeine was analysed using the same HPLC device. The peak area reading was 750.

Using the calibration curve, state the concentration of caffeine in the sample. 1 mark

The structure of caffeine is shown below.



d. The retention time for caffeine in the sample of coffee was 3 minutes in this HPLC instrument. The other components in the sample of coffee all had shorter retention times.

Explain, using the structure of caffeine, why it would have had the longest retention time. 2 marks

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e. In a **separate** HPLC instrument, the retention time of caffeine could not be distinguished as another component had a very similar retention time.

What change could be made to the instrument to ensure that the components had clearly defined retention times? Explain why this change would be effective. 2 marks

End of Section B

End of examination questions

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Name: _____

Multiple-Choice Answer Sheet

Section A

Question	Answer			
1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

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